Ghosh's Generalized Perfect q-Brick

Existence, Parametrizations, Geometry, and Interdisciplinary Interpretations

Soumadeep Ghosh

Kolkata, India

Abstract

The classical Perfect Euler Brick problem asks whether there exists a rectangular cuboid whose edges, face diagonals, and space diagonal are all integers. Motivated by this famous unsolved problem, we introduce a generalized continuous framework called *Ghosh's Generalized Perfect q-Brick*. For arbitrary real $q > 1$, we define a family of generalized edge lengths and diagonal lengths satisfying generalized Pythagorean relations in the L_q norm. We prove that such objects always exist, provide complete parametrizations, construct explicit examples, and discuss interpretations across mathematics, physics, economics, computer science, artificial intelligence, warfare economics, and complex systems theory.

1 Introduction

The classical Euler brick satisfies

$$a^2 + b^2 = d_1^2,$$

$$b^2 + c^2 = d_2^2,$$

$$c^2 + a^2 = d_3^2.$$

A perfect Euler brick additionally requires

$$a^2 + b^2 + c^2 = D^2.$$

Whether such an object exists remains an open problem.

We generalize the framework by replacing the Euclidean exponent 2 with an arbitrary real exponent $q > 1$.

2 Definition

Definition 1 (Ghosh's Generalized Perfect q-Brick)**.** *For real $q > 1$, a generalized perfect q-brick is a collection of positive functions*

$$a(q), b(q), c(q), d_1(q), d_2(q), d_3(q), D(q)$$

satisfying

$$a(q)^q + b(q)^q = d_1(q)^q,$$

$$b(q)^q + c(q)^q = d_2(q)^q,$$

$$c(q)^q + a(q)^q = d_3(q)^q,$$

and

$$a(q)^q + b(q)^q + c(q)^q = D(q)^q.$$

3 Existence Theorem

Define

$$A(q) = a(q)^q,$$

$$B(q) = b(q)^q,$$

$$C(q) = c(q)^q.$$

Then

$$a(q) = A(q)^{1/q},$$

$$b(q) = B(q)^{1/q},$$

$$c(q) = C(q)^{1/q}.$$

Theorem 1 (Universal Existence)**.** *Let*

$$A(q) > 0,$$

$$B(q) > 0,$$

$$C(q) > 0.$$

Define

$$d_1(q) = (A(q) + B(q))^{1/q},$$

$$d_2(q) = (B(q) + C(q))^{1/q},$$

$$d_3(q) = (C(q) + A(q))^{1/q},$$

and

$$D(q) = (A(q) + B(q) + C(q))^{1/q}.$$

Then the resulting system forms a generalized perfect q-brick.

Proof. Observe

$$d_1(q)^q = A(q) + B(q) = a(q)^q + b(q)^q.$$

Similarly,

$$d_2(q)^q = b(q)^q + c(q)^q,$$

and

$$d_3(q)^q = c(q)^q + a(q)^q.$$

Furthermore,

$$D(q)^q = A(q) + B(q) + C(q) = a(q)^q + b(q)^q + c(q)^q.$$

Therefore all defining equations hold.

□

4 Explicit Polynomial Family

Choose

$$A(q) = q,$$

$$B(q) = q + 1,$$

$$C(q) = q + 2.$$

Then

$$a(q) = q^{1/q},$$

$$b(q) = (q + 1)^{1/q},$$

$$c(q) = (q + 2)^{1/q}.$$

The generalized face diagonals become

$$d_1(q) = (2q + 1)^{1/q},$$

$$d_2(q) = (2q + 3)^{1/q},$$

$$d_3(q) = (2q + 2)^{1/q}.$$

The generalized body diagonal becomes

$$D(q) = (3q + 3)^{1/q}.$$

4.1 Example at $q = 3$

Quantity	Value
a	$\sqrt[3]{3}$
b	$\sqrt[3]{4}$
c	$\sqrt[3]{5}$
d_1	$\sqrt[3]{7}$
d_2	$\sqrt[3]{9}$
d_3	2
D	$\sqrt[3]{12}$

5 Exponential Family

Let

$$A(q) = e^{\alpha q},$$

$$B(q) = e^{\beta q},$$

$$C(q) = e^{\gamma q},$$

where

$$\alpha, \beta, \gamma > 0.$$

Then

$$a(q) = e^{\alpha},$$

$$b(q) = e^{\beta},$$

$$c(q) = e^{\gamma}.$$

The generalized diagonals become

$$d_1(q) = \left(e^{\alpha q} + e^{\beta q}\right)^{1/q},$$

$$d_2(q) = \left(e^{\beta q} + e^{\gamma q}\right)^{1/q},$$

$$d_3(q) = \left(e^{\gamma q} + e^{\alpha q}\right)^{1/q}.$$

6 Geometry

The generalized perfect q-brick naturally belongs to L_q geometry.

For a vector

$$x = (x_1, x_2, x_3),$$

the L_q norm is

$$|x|_q = \left(|x_1|^q + |x_2|^q + |x_3|^q\right)^{1/q}.$$

The generalized body diagonal is therefore

$$D(q) = |(a, b, c)|_q.$$

7 A Limit Theorem

Theorem 2. *For fixed positive constants*

$$a, b, c,$$

$$\lim_{q\to\infty} D(q) = \max(a, b, c).$$

Proof. A standard result from normed vector spaces states

$$\lim_{q\to\infty} \left(a^q + b^q + c^q\right)^{1/q} = \max(a, b, c).$$

Hence

$$\lim_{q\to\infty} D(q) = \max(a, b, c).$$

□

8 Computational Algorithm

Pseudo-Code

```
Input: q > 1

A = q
B = q + 1
C = q + 2

a  = A^(1/q)
b  = B^(1/q)
c  = C^(1/q)

d1 = (A+B)^(1/q)
d2 = (B+C)^(1/q)
d3 = (C+A)^(1/q)

D  = (A+B+C)^(1/q)

Return
(a,b,c,d1,d2,d3,D)
```

9 A Data Science Perspective

The generalized q-brick may be interpreted as a feature aggregation architecture.

Suppose

$$A(q), B(q), C(q)$$

represent feature strengths.

Then

$$D(q) = (A + B + C)^{1/q}$$

acts as a generalized aggregation operator.

As q increases, the system increasingly emphasizes dominant features.

This is analogous to attention mechanisms and max-pooling operators in machine learning.

10 Economic Interpretation

Suppose

$$A(q)$$

represents productive capital,

$$B(q)$$

represents labor,

and

$$C(q)$$

represents technology.

Then

$$D(q)$$

represents generalized productive capacity.

The exponent q controls substitutability between productive factors.

11 Warfare Economics Interpretation

Let

$$A(q)$$

represent military manpower,

$$B(q)$$

represent industrial capacity,

$$C(q)$$

represent logistics.
Then

$$D(q)$$

measures aggregate warfighting capability.
Large values of q correspond to systems dominated by their strongest component.
Small values correspond to more balanced force structures.

12 Psychological Interpretation

One may interpret

$$A(q)$$

as cognition,

$$B(q)$$

as emotion,

$$C(q)$$

as motivation.
Then

$$D(q)$$

becomes a generalized index of human functioning.

13 Biological Interpretation

Let

$$A(q)$$

represent metabolic capacity,

$$B(q)$$

represent immune function,

$$C(q)$$

represent reproductive fitness.
The generalized diagonal then measures aggregate biological performance.

14 Physics Interpretation

The generalized q-brick naturally appears in statistical physics and generalized metric spaces.

In particular,

$$D(q) = (a^q + b^q + c^q)^{1/q}$$

resembles generalized energy aggregation under non-Euclidean norms.

15 Chemistry Interpretation

Chemical systems often combine multiple reaction pathways.

The quantities

$$A(q), B(q), C(q)$$

may be interpreted as reaction intensities.

The generalized diagonal becomes a composite reaction-strength measure.

16 Political Science and International Relations

Consider

$$A(q) = \text{economic power},$$

$$B(q) = \text{military power},$$

$$C(q) = \text{technological power}.$$

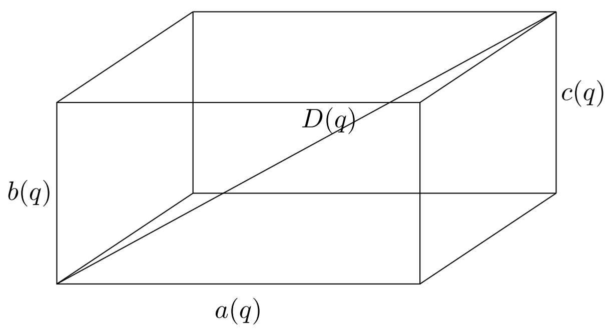
Then

$$D(q)$$

becomes a generalized national power index.

Different values of q correspond to different theories of power aggregation.

17 Illustrative Diagram



18 Conclusion

Unlike the classical Perfect Euler Brick problem, Ghosh's Generalized Perfect q-Brick always exists. Every choice of positive functions

$$A(q), B(q), C(q)$$

generates a valid generalized q-brick. Consequently, the construction defines an infinite-dimensional family of generalized metric objects whose applications extend far beyond geometry into economics, warfare analysis, machine learning, political science, biology, and complex systems.

References

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Glossary

q Generalized norm exponent.

L_q **Norm** Generalized distance measure.

$A(q), B(q), C(q)$ Underlying generating functions.

$a(q), b(q), c(q)$ Generalized edge lengths.

$d_1(q), d_2(q), d_3(q)$ Generalized face diagonals.

$D(q)$ Generalized body diagonal.

Universal Existence Property that every positive triple generates a valid q-brick.

Feature Aggregation Machine-learning interpretation of the generalized diagonal.